Google Developer Group
Bandung Institute of Technology

Team name	Tide Ledger
Project title	RapiLPJ: Pre-Submission Audit for Student Organization Activity Reports
Selected case	Case 1: Smarter Campus and Learning
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Abstract

1. Problem. Student treasurers must submit an activity accountability report (LPJ) after funded campus events before subsequent funds are disbursed. Faculty guidelines enforce disparate administrative rules: receipt seals, arithmetic balance, attendance ratios, revenue stamps, tax deductions, and deadlines. At Brawijaya Computer Science, incomplete reports halt upcoming proposals and extracurricular peer learning [6]. Treasurers lose valuable study hours to mechanical bookkeeping corrections and manual receipt assembly.

2. Intended Users. Treasurers and secretaries of Indonesian student organizations—study program associations, student executive boards (BEM), and clubs at Universitas Atma Jaya Makassar and peer institutions—who manage budgets without accounting training.

3. Supporting Observations or Evidence. Faculty research at Universitas Atma Jaya Makassar documented that the Student Affairs Bureau (BKAM) struggles with unstandardized reports and verification delays that threaten compliance [1]. Brawijaya Cultural Studies mandates stamped receipts, attendance lists per meal, and exact ledger balances [2]. UGM Economics enforces Rp10,000 revenue stamps on receipts of Rp5,000,000+ under Law 10/2020 and photocopies for thermal slips [3]. ITS Interior Design enforces a 10-day deadline [4], while ITS Biology requires separate line items and tax inclusion [5]. Student officers consistently struggle with tax classification and budget reconciliations [6].

4. Proposed Solution and Core User Flow. RapiLPJ couples an open-source browser audit engine with a governance co-pilot and automated PDF dossier compiler in three steps: (1) Deterministic Pre-Audit & Offline Persistence: A client-side JavaScript engine checks zero-tolerance rules—duplicate invoices, date bounds, revenue stamps, thermal fade warnings, and >10% budget variances—while persisting portable .lpj draft files. (2) Governance Advisory & Addendum Synthesis: On infractions, the co-pilot explains statutory rules and drafts formal Addendum letters via deterministic templates or an optional client-side Gemini 2.0 Flash API. (3) Auto-Constructed PDF Dossier: The treasurer exports a unified PDF combining reconciliation tables, addendum letters, an indexed receipt image appendix, and a SHA-256 integrity hash.

5. How It Differs from Existing Alternatives. Pure LLMs suffer from arithmetic hallucinations on balance sheets. Spreadsheets lack automated statutory checks, receipt image indexing, and addendum drafting. Campus portals merely collect finished files without pre-submission triage. RapiLPJ enforces a strict neuro-symbolic boundary: financial arithmetic and compliance verification run exclusively via open-source deterministic algorithms in browser memory, while generative AI is strictly isolated to optional advisory and addendum prose refinement.

6. Planned Technologies. A privacy-preserving client-side stack. Client Engine: HTML5, CSS3 (bilingual EN/ID and light/dark accessibility), open-source JavaScript rule evaluation, FileReader receipt image indexing, .lpj JSON state serialization, and SHA-256 attestation. Static Hosting: Vercel CDN without backend database storage, ensuring financial confidentiality. Advisory Layer: Built-in deterministic statutory templates with optional client-side Google Gemini 2.0 Flash API integration.

7. Expected Benefit and Development Plan. Expected benefit: eliminates mechanical LPJ rejections, accelerates university funding disbursements, saves study hours, and teaches experiential fiscal governance to student leaders. Plan: Semifinal phase validates the engine with 30 treasurers at Universitas Atma Jaya Makassar and expands faculty rule presets; Final Day sprint builds multi-page receipt OCR batch extraction and QR-verifiable advisor sign-off sheets. Institutional rule changes are mitigated via configurable policy toggles.

Source Acknowledgments. [1] P. Siada & A. L. Tungadi, TEMATIKA J. Penelit. Tek. Inform. Sist. Inf. UAJM, 2021, doi:10.56963/tematika.v9i1.119. [2] Pedoman LPJ, FIB Universitas Brawijaya, fib.ub.ac.id. [3] Prosedur Proposal & LPJ SIPRO, FEB UGM & UU Bea Meterai No. 10/2020. [4] SOP HIMA, Desain Interior ITS, its.ac.id. [5] SOP LPJ Mahasiswa Biologi, ITS, its.ac.id. [6] Surat Edaran Mekanisme LPJ Ormawa, FILKOM UB, 2026. AI disclosure: Claude and Gemini supported research, deterministic engine code, and drafting.